In class lab #3 for Numerical Analysis

Your name: _____

scan and turn the assignment before 16:00

Part I. Newton's Method We remember that Taylor's formula:

$$f(x) = f(x_0)+f'(x_0)(x-x_0)+\frac{f''(x_0)}{2!}(x-x_0)^2+\frac{f'''(x_0)}{3!}(x-x_0)^3+\cdots+\frac{f^{(k)}(x_0)}{k!}(x-x_0)^k+\frac{f^{(k+1)}(c)}{(k+1)!}(x-x_0)^{k+1}. \tag{1}$$

Using the first two terms of the right-hand-side of the above equation, we obtain

$$x = x_0 - \frac{f(x_0)}{f'(x_0)}. \tag{2}$$

Solving for x gives an approximation for the root, which we call x_1. Next, the entire process is repeated, beginning with x_1, to produce x_2, and so on, yielding the following iterative formula:

$$x_0 = \textbf{intial guess}$$

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)} \text{ for } i = 0, 1, 2, \cdots .$$

Indeed, **the Newton's method is a kind of the Fixed-Point iteration method**. In fact, we let

$$g(x) = x - \frac{f(x)}{f'(x)}, \tag{3}$$

and we find the root r is a fixed point of the function $g(x)$.

Problem 1. (20 points) Creating the `newton.m` function to solve the following equations with $x_0 = 1$ within $tol = 10^{-10}$. maxIt can be set as 1000. Answer with the solution and the number of iterations.
(a) $x^4 - 3x^2 - 6 = 0$ (b) $x^3 = 2 - 2x + \ sinx$

Problem 2. (30 points) Consider $f(x) = 4x^4 - 6x^2 - 11/4$.

(a) Plot the graph of $f(x)$.
(b) Apply Newtons Method to $f(x)$ with starting guess $x_0 = 1/2$. What happened? State what you observed.
(c) Apply Newtons method to $f(x)$ and find **positive initial guesses** that converge to the positive root and the negative root respectively.

Problem3. (20 points) Suppose Newtons Method is applied to the function $f(x) = \frac{1}{x}$. If the initial guess is $x_0 = 1$, find x_{100} without running the calculation.

Part II. Secant Method The Secant Method is similar to the Newton's Method, but replaces the derivative by a difference quotient. An approximation for the derivative at the current guess x_i is the difference quotient

$$\frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}}$$

A straight replacement of this approximation for $f'(x_i)$ in Newton's Method yields the Secant Method.

$$x_0, x_1 = \textbf{ initial guesses}$$

$$x_{i+1} = x_i - \frac{f(x_i)(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})} \text{ for } i = 1, 2, 3, \ldots$$

Unlike Fixed-Point Iteration and Newtons Method, **two starting guesses** are needed to begin the Secant Method.

Problem 4. (20 points) Creating the `secant.m` function to solve the following equations with $x_0 = 1$ and $x_1 = 2$ within $tol = 10^{-5}$. maxIt can be set as 1000. Answer with the solution and the number of iterations.
(a) $x^4 - \ lnx = 2$ (b) $e^x + \sin x = 4$

Problem 5. (10 points)

(1) Type "help fzero" in the command window and answer the function of fzero.

(2) Please use `fzero` to obtain the solution near $x = 1$ of (a) $x^3 = 2x + 2$ (b) $e^x + x = 7$ (c) $e^x + \sin(x) = 4$.